

ps1

September 21, 2025

1 Econ 141 – Problem Set 1

In this problem set we will be doing t-tests, among other things. There are many ways to do t-testing in Python; we will be using a package called `scipy`. More specifically, `scipy`'s `stats` tools. The method that does t-testing of means for two independent samples is `ttest_ind`. [Here](#) is the documentation. The documentation for many of the Python packages we will be using is very helpful and you are encouraged to reference it. It will almost always be the fastest way to answer questions about the usage of these packages. We'll go through an example of how to use `ttest_ind`.

Suppose we have a dataset called `coinflips` that has two columns of observations, one called `coin1`, the other `coin2`. Each entry in each column is either a 1 or 0 depending on the result of the coin for a particular flip. If we want to do a t-test for equality of means of the coins (which in this case is also testing if the coins have the same chance of landing on heads) we would do the following:

```
stats.ttest_ind(coinflips['coin1'], coinflips['coin2'])
```

which would return something like this: `Ttest_indResult(statistic=-2.053, pvalue=0.041)`. These results might indicate that the first coin is probably less likely to land on heads than the second coin. This is because the first argument to `ttest_ind` was `coin1`, and the t-stat is negative. If the t-stat were positive, it might indicate that the first coin is more likely to land on heads. To more clearly see the results, we can append `.statistic` at the end of the `ttest_ind` command to just get `-2.053`, and `.pvalue` to just get `0.041`.

By default, `ttest_ind` assumes a two-sided test. It also assumes equal variance between the two samples. For unequal variance, we can do

```
stats.ttest_ind(coinflips['coin1'], coinflips['coin2'], equal_var=False)
```

If our data have missing values (we'll see an example of this later in the problem set) `ttest_ind` by default will not give us the results we want (a t-test ignoring the missing values). To tell `scipy` to only consider non-missing values, we do

```
stats.ttest_ind(coinflips['coin1'], coinflips['coin2'], nan_policy='omit')
```

Before getting started on the assignment, run the cell below which will import the packages we need.

Important: As mentioned in problem set 0, if you leave this notebook alone for a while and come back, to save memory datahub will “forget” which code cells you have run, and you may need to restart your kernel and run all of the cells from the top. That includes the code cell below that

imports packages. If you get <something> not defined errors, this is because you didn't run an earlier code cell that you needed to run.

```
[4]: import numpy as np
import pandas as pd
from scipy import stats
import matplotlib.pyplot as plt
```

1.1 Problem 1. Golden State Warriors

The 2015-16 season was historic for Stephen Curry and the Golden State Warriors. They finished the regular season with a 73-9 record, which beat out the Chicago Bulls 1995-96 record of 72-10 for the best regular season record in NBA history. This question takes a statistical look at the business of pro basketball. The accompanying dataset `nba.csv` contains information on 269 NBA players for one specific season. Here is a description of some of the variables in the dataset:

Variable	Description
wage	player's salary for the season in thousands of nominal dollars
exper	number of years in the league
age	age of the player
coll	years of college completed
games	number of games played that season
minutes	total number of minutes player in the season
guard	dummy indicator whether player plays guard position
forward	dummy indicator whether player plays forward position
center	dummy indicator whether player plays center position
points	season average number of points per game played
rebounds	season average number of rebounds per game played
assists	season average number of assists per game played

```
[2]: # Don't forget to run this cell that reads in the data
nba = pd.read_csv("nba.csv")
nba.head()
```

```
[2]: married    wage  exper  age  coll  games  minutes  guard  forward  center  \
0           1  1002.5     4   27     4     77     2867     1         0         0
1           1  2030.0     5   28     4     78     2789     1         0         0
2           0   650.0     1   25     4     74     1149     0         0         1
3           0  2030.0     5   28     4     47     1178     0         1         0
4           0   755.0     3   24     4     82     2096     1         0         0

    points  rebounds  assists  draft  allstar  avgmin  black  children
0        16         4        5   19.0         0   37.23     1         0
1        13         3        9   28.0         0   35.76     1         1
2         6         3        0   19.0         0   15.53     1         0
3         7         5        2    1.0         0   25.06     1         0
4        11         4        3   24.0         0   25.56     1         0
```

Question 1.a. To begin with, test whether players who play the guard position are paid the same as other players. Be sure to report the results of your test including the t-statistic and p-value.

This question is for your code, the next is for your explanation.

Hint: For those unfamiliar with American basketball, players are classified as playing one of three positions: guard, forward and center.

```
[3]: nba_guard = nba[nba['guard']==1]['wage']
nba_not_guard = nba[nba['guard']==0]['wage']

ttest_1a = stats.ttest_ind(nba_guard, nba_not_guard)

tstat_1a = ttest_1a.statistic
pval_1a = ttest_1a.pvalue

print("t-stat: {}".format(tstat_1a))
print("p-value: {}".format(pval_1a))
```

```
t-stat: -2.0530342128806582
p-value: 0.041043637620105405
```

Question 1.b. Explain.

After running that code, we can get t and p, if $p < 0.05$, then we can come into a conclusion that the wage differences between nba_guard and nba_not_guard shows a huge difference, and vice versa. So the wage differences between nba_guard and nba_not_guard shows a huge difference.

Question 1.c. Do NBA players who complete college degree get paid more or less than those who do not? Test this hypothesis. Explain your results.

This question is for your code, the next is for your explanation.

Hint: Define a new variable degree to indicate whether the player completed 4 or more years of college.

```
[4]: nba['degree'] = nba['coll']>=4

nba_degree = nba[nba['degree']==True]['wage']
nba_no_degree = nba[nba['degree']==False]['wage']

ttest_1c = stats.ttest_ind(nba_degree, nba_no_degree)

tstat_1c = ttest_1c.statistic
pval_1c = ttest_1c.pvalue

print("t-stat: {}".format(tstat_1c))
print("p-value: {}".format(pval_1c))
```

```
t-stat: -2.3705333284217867
p-value: 0.0184730432847849
```

Question 1.d. Explain.

We use degree to test whether the nba player has gone to the college and to get t and p values to test, if $p < 0.05$, we could come into a conclusion that the wages have differences due to the college degrees. Therefore, we can say that the wages have differences due to the college degrees.

Question 1.e. Compute the *productivity* of each player in terms of the average number of points scored per minute of playing time. Note that the variable points is itself an average per game for the sampled season. Test whether guards are as productive as players who play other positions in this sense.

This question is for your code, the next is for your explanation.

```
[5]: nba['productivity'] = nba['points'] / nba['minutes']

nba_guard = nba[nba['guard'] == 1]['productivity']
nba_not_guard = nba[nba['guard'] == 0]['productivity']

ttest_1e = stats.ttest_ind(nba_guard, nba_not_guard)

tstat_1e = ttest_1e.statistic
pval_1e = ttest_1e.pvalue

print("t-stat: {}".format(tstat_1e))
print("p-value: {}".format(pval_1e))
```

```
t-stat: -1.7767355680677752
p-value: 0.07675030716366808
```

Question 1.f. Explain.

From our test, $p > 0.05$, we can say that guards are as productive as players who play other positions in this sense.

Question 1.g. Players do more on the court than just put the ball in the hoop. They also rebound the ball and assist other players. Data on these two measures are given as a per-game average alongside points. Find the sample correlations between the three performance variables: points, rebounds, and assists.

Hint: The `.corr()` command is useful here.

```
[7]: nba[['points', 'assists', 'rebounds']].corr()
```

```
[7]:
```

	points	assists	rebounds
points	1.000000	0.539269	0.563324
assists	0.539269	1.000000	0.059956
rebounds	0.563324	0.059956	1.000000

Question 1.h. To take all the performance measures into account, create a performance index as a weighted sum of the three measures: $\text{index} = \text{points} + \text{rebounds} + 2 \cdot \text{assists}$. Using this index, test whether guards have the same performance as players at other positions.

This question is for your code, the next is for your explanation.

```
[11]: nba['index'] = nba['points']+nba['rebounds']+2*nba['assists']

nba_guard = nba[nba['guard']==1]['index']
nba_not_guard = nba[nba['guard']==0]['index']

ttest_1h = stats.ttest_ind(nba_guard, nba_not_guard)

tstat_1h = ttest_1h.statistic
pval_1h = ttest_1h.pvalue

print("t-stat: {}".format(tstat_1h))
print("p-value: {}".format(pval_1h))
```

t-stat: 2.1909453801070837

p-value: 0.029320202223591406

Question 1.i. Explain.

If $p > 0.05$, we can say that guards have the same performance as players at other positions. p value is 0.03, so guards have different performance as players at other positions

Question 1.j. Finally, NBA general managers are very interested to know whether they are getting their money's worth, so want to know whether players are over or under paid given their performance. Compute a variable equal to the performance index per \$1,000 of salary and again test whether guards are paid the same relative to performance as other positions.

This question is for your code, the next is for your explanation.

```
[12]: nba['payoff'] = nba['points']/nba['wage']

nba_guard = nba[nba['guard']==1]['payoff']
nba_not_guard = nba[nba['guard']==0]['payoff']

ttest_1j = stats.ttest_ind(nba_guard, nba_not_guard)

tstat_1j = ttest_1j.statistic
pval_1j = ttest_1j.pvalue

print("t-stat: {}".format(tstat_1j))
print("p-value: {}".format(pval_1j))
```

t-stat: 2.213787185959181

p-value: 0.0276886991956593

Question 1.k. Explain.

The $p < 0.05$ shows that guards are paid the different relative to performance as other positions.

1.2 Problem 2. Crime on Campus

Crime on campus has become a critical national policy issue. This question explores some data collected about crime on a random sample of U.S. colleges and universities in 2001. The accompanying dataset `crime.csv` contains the following variables:

Variable	Description
enrollment	number of full-time enrolled undergrad and grad students
private	1 if the school is private, and 0 if the school is public
police	number of full-time equivalent campus police officers
crime	number of confirmed on-campus crimes per year

```
[5]: crime = pd.read_csv("crime.csv")
      crime.head()
```

```
[5]:  enrollment  private  police  crime
0         11967         0       11     89
1         12412         0        8    140
2          4100         0        6     53
3         17812         0       16    395
4         16798         0       15    252
```

```
[3]: # This code cell is for question 2.a.
```

Question 2.a. Complete the following table of summary statistics. A code cell has been provided above for you to do your work for this question if you need it. An outline of the table is provided below. Replace the ... with actual numbers.

Hint: The command `.describe()` will be useful. For example, if your dataset is called `nba`, the usage is `nba.describe()`.

	Enrollment	Police	Crime
Number of observations	...	...	...
Sample mean	...	...	...
Sample median	...	...	...
Sample standard deviation	...	...	...
Sample mean for public schools	...	...	...
Sample mean for private schools	...	...	...

Question 2.b. Compute the sample correlation between enrollment, police, and crime. Do the values you find make sense?

```
[15]: crime[['enrollment', 'police', 'crime']].corr()
```

```
[15]:  enrollment  police  crime
enrollment    1.000000  0.715053  0.836044
police         0.715053  1.000000  0.723310
```

```
crime          0.836044  0.723310  1.000000
```

Question 2.c. Do the values you find above make sense?

Yes, it makes sense. We can see the correlation efficient between enrollment and police is 0.836044. We can explain it as the higher enrollment means higher population mobility, asking schools to pay more attention to the safety issue and hire more policemen.

Question 2.d. Test the hypothesis that the crime levels are the same in private and public schools by performing a t-test for equality of means of two subsamples. Is there a difference at the 5% significance level? At the 1% level? Do your conclusions depend on whether you assume the same and different variances for the two types of schools? Explain.

This question is for your code, the next is for your explanation.

```
[7]: crime_public = crime[crime['private']==0]['crime']
     crime_private = crime[crime['private']==1]['crime']

     ttest_2d_unequal_var = stats.ttest_ind(crime_public, crime_private,
     ↪equal_var=False)

     tstat_2d_unequal = ttest_2d_unequal_var.statistic
     pval_2d_unequal = ttest_2d_unequal_var.pvalue

     ttest_2d_equal_var = stats.ttest_ind(tstat_2d_unequal, pval_2d_unequal)

     tstat_2d_equal = ttest_2d_equal_var.statistic
     pval_2d_equal = ttest_2d_equal_var.pvalue

     print("t-stat unequal variance: {}".format(tstat_2d_unequal))
     print("p-value unequal variance: {}".format(pval_2d_unequal))
     print("t-stat equal variance: {}".format(tstat_2d_equal))
     print("p-value equal variance: {}".format(pval_2d_equal))
```

```
t-stat unequal variance: 4.8504897937789995
p-value unequal variance: 8.356792319851925e-06
t-stat equal variance: nan
p-value equal variance: nan
```

Question 2.e. Explain.

Calculated p value is much smaller than both 0.05 and 0.01, so we reject the null hypothesis that crime levels are the same in private and public schools. There is a significant difference.

Question 2.f. Since it is likely that more crimes occur on bigger campuses, generate a new variable called “crimerate,” defined as the number of crimes per 1,000 students. Test whether private and public schools have different crime rates (allowing for potentially unequal variances). Is there a difference at the 5% level? At the 1% level? Do your conclusions depend on whether you assume the same and different variances for the two types of schools? Explain.

This question is for your code, the next is for your explanation.

```
[28]: crime['crimerate'] = crime['crime']/(crime['enrollment'])
      #the number of crimes per 1,000 students

      crime_public = crime[crime['private']==0]['crime']
      crime_private = crime[crime['private']==1]['crime']
      #Test whether private and public schools have different crime rates (allowing
      ↪for potentially unequal variances)
      ttest_2f_unequal_var = stats.ttest_ind(crime_public, crime_private,
      ↪equal_var=False)

      tstat_2f_unequal = ttest_2f_unequal_var.statistic
      pval_2f_unequal = ttest_2f_unequal_var.pvalue

      ttest_2f_equal_var = stats.ttest_ind(tstat_2f_unequal, pval_2f_unequal)

      tstat_2f_equal = ttest_2f_equal_var.statistic
      pval_2f_equal = ttest_2f_equal_var.pvalue

      print("t-stat unequal variance: {}".format(tstat_2f_unequal))
      print("p-value unequal variance: {}".format(pval_2f_unequal))
      print("t-stat equal variance: {}".format(tstat_2f_equal))
      print("p-value equal variance: {}".format(pval_2f_equal))
```

```
t-stat unequal variance: 4.8504897937789995
p-value unequal variance: 8.356792319851925e-06
t-stat equal variance: nan
p-value equal variance: nan
```

Question 2.g. Explain.

Calculated p-value is smaller than 0.05 and 0.01, so we reject the hypothesis that private and public schools have different crime rates, means that the crime rates does not depend on the private or public character of the schools.

1.3 Problem 3. Post Katrina employment in New Orleans

The following table has information about home location and employment status of New Orleans residents taken from the U.S. Current Population Survey in August 2006, a year after hurricane Katrina inflicted severe damage on the Gulf Coast. Observations are made on 249 randomly selected survey respondents who had to evacuate their home due to the hurricane. The table entries are the number of survey respondents.

	Employed	Unemployed	Total
Returned to pre-storm address	139	8	147
Have not yet returned	79	23	102
Total	218	31	249

Question 3.a. Create another table with the joint and marginal probabilities associated with this sample. Create another table with the conditional distribution of employment status given whether or not the resident has returned to their home. An outline of the tables is provided below. Replace the ... with actual numbers. Make sure to show your work for how you derived the tables!

You may either: * Write your math in the cell below using LaTeX typesetting (recommended).
 * Write your math on a tablet and include the exported pdf with the pdf submission of this assignment.
 * Write your math on paper and include a scan with the pdf submission of this assignment.

Joint	Employed	Unemployed	Marginal
Returned to pre-storm address	...	...	...
Have not yet returned	...	...	...
Marginal	...	...	...

Conditional	Employed	Unemployed
Given returned to pre-storm address	...	...
Given have not yet returned	...	...

	Employed	Unemployed	Marginal
Returned	$\frac{139}{249} \approx 0.558$	$\frac{8}{249} \approx 0.032$	$\frac{147}{249} \approx 0.590$
Not Returned	$\frac{79}{249} \approx 0.317$	$\frac{23}{249} \approx 0.092$	$\frac{102}{249} \approx 0.410$
Marginal	$\frac{218}{249} \approx 0.875$	$\frac{31}{249} \approx 0.125$	$\frac{249}{249} = 1$

Conditional	Employed	Unemployed
Given Returned	$\frac{139}{147} \approx 0.946$	$\frac{8}{147} \approx 0.054$
Given Not Returned	$\frac{79}{102} \approx 0.775$	$\frac{23}{102} \approx 0.225$

Question 3.b. Using this last table find the expectation of a resident being employed, conditional on returning to their home. To do this, assign values to the two variables: 1 = returned to home and 0 = did not; 1 = employed and 0 = unemployed. Using the same table, confirm the law of iterated expectations.

You may either: * Write your math in the cell below using LaTeX typesetting (recommended).
 * Write your math on a tablet and include the exported pdf with the pdf submission of this assignment.
 * Write your math on paper and include a scan with the pdf submission of this assignment.

The expectation of a resident being employed is expressed as $P(\text{Employed} | \text{Returned})$. The joint table shows the number of people who are employed and returned is 139. The total number of people who returned is 147. Therefore, the conditional probability is: $P(\text{Employed} | \text{Returned}) = \frac{\text{Number of Employed and Returned}}{\text{Total number of Returned}} = \frac{139}{147}$ Assigning values: 1 = Returned to home 0 = Did not return 1 = Employed 0 = Unemployed Then, let X be the variable for returning status ($X = 1$ for

returned, $X = 0$ for not returned) and Y be the variable for employment status ($Y = 1$ for employed, $Y = 0$ for unemployed). The expectation of a resident being employed, conditional on returning to their home is calculated as the sum of each possible employment status value multiplied by its conditional probability.

$$E[Y|X = 1] = \sum_{y \in \{0,1\}} y \cdot P(Y = y|X = 1)$$

$$E[Y|X = 1] = (1) \cdot P(Y = 1|X = 1) + (0) \cdot P(Y = 0|X = 1)$$

$$P(Y = 1|X = 1) = \frac{139}{147}$$

$$P(Y = 0|X = 1) = \frac{8}{147}$$

$$E[Y|X = 1] = (1) \cdot \frac{139}{147} + (0) \cdot \frac{8}{147} = \frac{139}{147} \approx 0.9456$$

The law of iterated expectations states that $E[Y] = E[E[Y|X]]$. In this question, it can be expressed as:

$$E[Y] = E[Y|X = 1] \cdot P(X = 1) + E[Y|X = 0] \cdot P(X = 0)$$

The total number of employed residents is 218, and the total number of survey respondents is 249.

$$E[Y] = P(Y = 1) = \frac{218}{249} \approx 0.8755$$

- $E[Y|X = 1]$: The expectation of being employed, given that they returned. We already calculated this as $\frac{139}{147} \approx 0.9456$.
- $E[Y|X = 0]$: The expectation of being employed, given that they did not return. The number of employed residents who did not return is 79, and the total number of people who did not return is 102.

$$E[Y|X = 0] = P(Y = 1|X = 0) = \frac{79}{102} \approx 0.7745$$

- $P(X = 1)$: The probability of returning. The total number of people who returned is 147, and the total respondents are 249.

$$P(X = 1) = \frac{147}{249} \approx 0.5904$$

- $P(X = 0)$: The probability of not returning. The total number of people who did not return is 102, and the total respondents are 249.

$$P(X = 0) = \frac{102}{249} \approx 0.4096$$

Finally, we can plug these values back into the law of iterated expectations formula to confirm that it holds:

$$E[Y] = E[Y|X = 1] \cdot P(X = 1) + E[Y|X = 0] \cdot P(X = 0)$$

$$E[Y] = \left(\frac{139}{147}\right) \cdot \left(\frac{147}{249}\right) + \left(\frac{79}{102}\right) \cdot \left(\frac{102}{249}\right)$$

$$E[Y] = \frac{139}{249} + \frac{79}{249} = \frac{218}{249} \approx 0.8755$$

Since the calculation using the law of iterated expectations matches the unconditional expectation, the law is confirmed.

Question 3.c. Compute the sample covariance of return status and employment status.

You may either: * Write your math in the cell below using LaTeX typesetting (recommended). * Write your math on a tablet and include the exported pdf with the pdf submission of this assignment. * Write your math on paper and include a scan with the pdf submission of this assignment.

Assume X be the variable for return status, where $X = 1$ if the resident returned to their pre-storm address and $X = 0$ if they have not yet returned. Let Y be the variable for employment status, where $Y = 1$ if the resident is employed and $Y = 0$ if they are unemployed.

$$Cov(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

where: - n is the total number of observations (249). - x_i and y_i are the values for each observation. - $\bar{x}$ is the sample mean of X . - $\bar{y}$ is the sample mean of Y .

The mean of the return status:

$$\bar{x} = P(X = 1) = \frac{\text{Number of people who returned}}{\text{Total number of people}} = \frac{147}{249} \approx 0.5904$$

$$\bar{y} = P(Y = 1) = \frac{\text{Number of employed people}}{\text{Total number of people}} = \frac{218}{249} \approx 0.8755$$

the sum of the products of the deviations:

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

, where $E[X] = \bar{x}$ and $E[Y] = \bar{y}$. X and Y can only be 0 or 1, the product XY is only non-zero when both $X = 1$ and $Y = 1$.

$$E[XY] = P(X = 1, Y = 1) = \frac{139}{249} \approx 0.5582$$

Put it into the equation,

$$Cov(X, Y) = E[XY] - \bar{x}\bar{y}$$

$$Cov(X, Y) = \frac{139}{249} - \left(\frac{147}{249}\right)\left(\frac{218}{249}\right)$$

$$Cov(X, Y) \approx 0.5582 - (0.5904)(0.8755)$$

$$Cov(X, Y) \approx 0.5582 - 0.5169$$

$$Cov(X, Y) \approx 0.0413$$

The positive covariance suggests a positive linear relationship between returning home and being employed. This means that a resident who returned to their home is more likely to be employed, and vice versa.

Question 3.d. Is current employment status statistically independent of the return status? Along with part 3.c, what does this say about the relationship between return status and employment status? Justify your answer.

The covariance between X and Y is 0.0413, belongs to (-1,1), and if X and Y is independent, the covariance need to be 0, meaning that they are not independent.

Question 3.e. Give two plausible reasons that could explain the difference in employment status between the residents who returned to their homes and those who did not.

Firstly, from a economic perspective, residents who had stable jobs and income were likely in a better financial position to rebuild or repair their homes, making it more feasible for them to return. People who were unemployed or had unstable jobs would have faced greater difficulty in affording the return. Secondly, the employment location may influence the decision of people's return places. It can be seen that many employed residents worked in the same area as their pre-storm homes, which may also contribute to the return choices.

1.4 Problem 4. Pollution

For this question, we will use a dataset on pollution. It contains information on the gross domestic product (GDP) and yearly CO2 emissions of 214 countries for the year 2010. You will find that the variable names clearly indicate what they measure, e.g. `co2pc` is CO2 per capita. Notice there are some missing values, which you can see from the `NaN` (not a number) entries in some places in the data.

```
[8]: pollution = pd.read_csv("pollution.csv")
      pollution.head()
```

```
[8]:   year      countryname countrycode      gdp      gdppc      co2 \
0  2010          Zambia          ZMB  9.799629e+09  741.4421  2427.554
1  2010  French Polynesia          PYF         NaN         NaN   883.747
2  2010          Monaco          MCO         NaN         NaN         NaN
3  2010          Ukraine          UKR  9.057726e+10  1974.6212  304804.720
4  2010  Venezuela, RB          VEN  1.750000e+11  6010.0270  201747.340
```

```
      co2pc  population  oecd
0  0.183669   13216985   0.0
1  3.296764    268065   0.0
2      NaN     36845   0.0
3  6.644867   45870700   0.0
4  6.946437   29043283   0.0
```

```
[9]: # This code cell is for question 4.a.
      #oecd sample mean&deviation
      oecd_mem_gdp=pollution[pollution['oecd']==1.0]['gdppc']
      oecd_not_mem_gdp=pollution[pollution['oecd']==0.0]['gdppc']
      stats_gdppc = pollution.groupby('oecd')['gdppc'].agg(['mean', 'std']).
      ↪reset_index()
```

```
print("GDP per capita Statistics:")
print(stats_gdppc)
```

GDP per capita Statistics:

	oecd	mean	std
0	0.0	5585.894420	9827.862772
1	1.0	31423.908076	17180.878835

```
[10]: # This code cell is for question 4.a.
#co2pc sample mean&deviation
oecd_mem_co2=pollution[pollution['oecd']==1.0]['co2pc']
oecd_not_mem_co2=pollution[pollution['oecd']==0.0]['co2pc']
stats_co2pc = pollution.groupby('oecd')['co2pc'].agg(['mean', 'std']).
    ↪reset_index()
print("CO2 per capita Statistics:")
print(stats_co2pc)
```

CO2 per capita Statistics:

	oecd	mean	std
0	0.0	4.074291	6.430327
1	1.0	8.984034	4.099933

Question 4.a. The variable `oecd` is a dummy indicator of each country's membership in the Organization for Economic Cooperation and Development (OECD): 1 = a member in OECD, 0 = not a member. The OECD consists of several dozen of the largest, most developed economies in the world. Compare the sample mean and standard deviation of per-capita GDP between the OECD and non-OECD countries. Do the same with per-capita CO2 emissions. A code cell has been provided for you above to do your work for this question.

OECD countries have both higher mean and greater variability in GDP per capital and CO2 per capital compared to non-OECD countries, reflecting their more advanced and diverse economies.

Question 4.b. Conduct a t-test of whether the sample means of CO2 emissions per-capita are significantly different between the OECD and non-OECD countries. Did you choose to assume variances of the two groups were equal or unequal? Explain why.

This question is for your code, the next is for your explanation.

```
[15]: pollution_oecd = pollution[pollution['oecd']==1]['co2pc'].dropna()
pollution_no_oecd = pollution[pollution['oecd']==0]['co2pc'].dropna()
ttest_4b = stats.ttest_ind(pollution_oecd, pollution_no_oecd, equal_var=False,
    ↪nan_policy='omit')

tstat_4b = ttest_4b.statistic
pval_4b = ttest_4b.pvalue

print("t-stat: {}".format(tstat_4b))
print("p-value: {}".format(pval_4b))
```

```
t-stat: 5.682465161974536
p-value: 2.683581293169389e-07
```

Question 4.c. Explain.

Assume the variances of the 2 groups were unequal because there may be some differences between the real condition in OECD countries and non-OECD countries.

In the remaining parts, we will explore the relationship between GDP and CO2 emissions. This relationship has also been called the “Environmental Kuznets Curve.”

Question 4.d. Approximate the growth rates of GDP and CO2 by first generating variables that are the natural logarithms of the two variables. Why would we examine the growth rates instead of the absolute levels of emissions and GDP? The function `np.log()` will be helpful. It can take a column as an argument in the parentheses. For example, if we wanted to take the natural log of a number instead of a column we would do `np.log(10)`. `np` is the shortcut for `numpy`, which is another useful package for doing math.

This question is for your code, the next is for your explanation.

```
[16]: pollution['log_gdp'] = np.log(pollution['gdp'])
      pollution['log_co2'] = np.log(pollution['co2'])

      pollution.head()
```

```
[16]:
```

	year	countryname	countrycode	gdp	gdppc	co2	\
0	2010	Zambia	ZMB	9.799629e+09	741.4421	2427.554	
1	2010	French Polynesia	PYF	NaN	NaN	883.747	
2	2010	Monaco	MCO	NaN	NaN	NaN	
3	2010	Ukraine	UKR	9.057726e+10	1974.6212	304804.720	
4	2010	Venezuela, RB	VEN	1.750000e+11	6010.0270	201747.340	

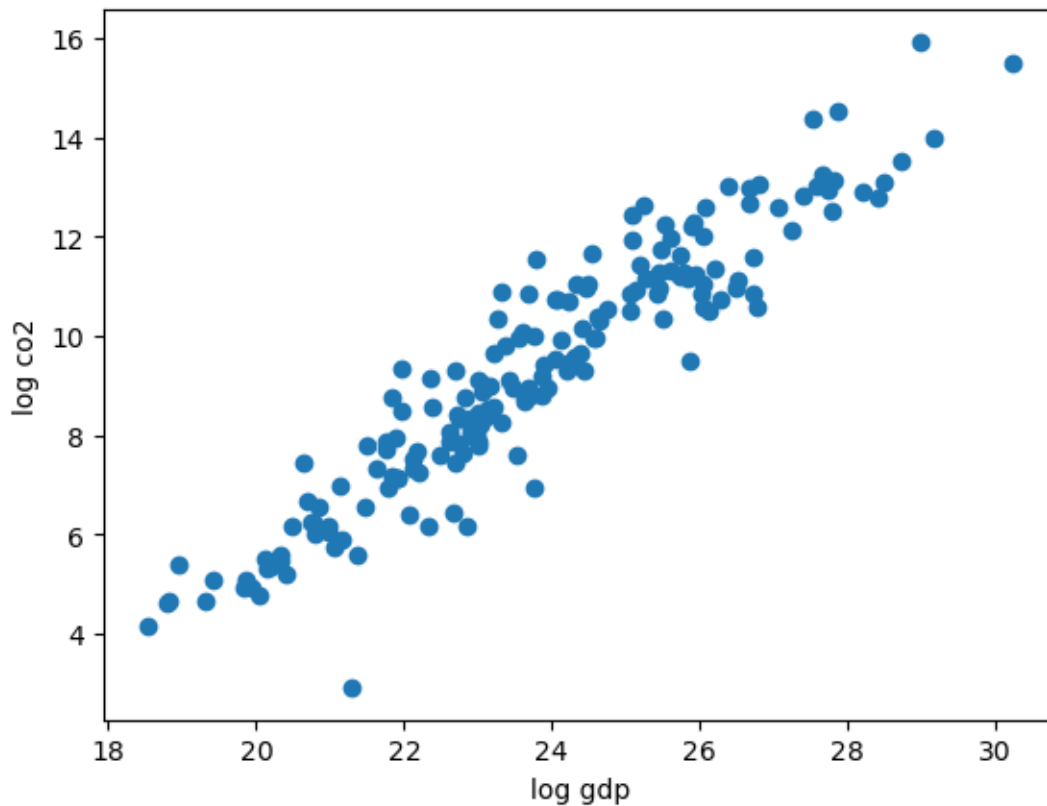
	co2pc	population	oecd	log_gdp	log_co2
0	0.183669	13216985	0.0	23.005610	7.794639
1	3.296764	268065	0.0	NaN	6.784171
2	NaN	36845	0.0	NaN	NaN
3	6.644867	45870700	0.0	25.229469	12.627427
4	6.946437	29043283	0.0	25.888052	12.214771

Question 4.e. Explain.

Growth rate reflects proportional changes in GDP and CO₂, which better reflect economic and environmental trends than absolute differences, and log-transforming converts exponential growth to linear growth, making it easier to see and analyze trends and relationships.

Below is code that generates a scatterplot of the two variables created in part 4.d. Run it. `plt.scatter()` takes in two columns from our data and generates a scatterplot. The first column you give it corresponds to the x-axis, the second column the y-axis. `plt.xlabel()` and `plt.ylabel()` just give names to the axes. The `;` at the end of the code is just there to suppress an output that isn't really helpful.

```
[17]: plt.scatter(pollution['log_gdp'], pollution['log_co2'])
plt.xlabel("log gdp")
plt.ylabel("log co2");
```

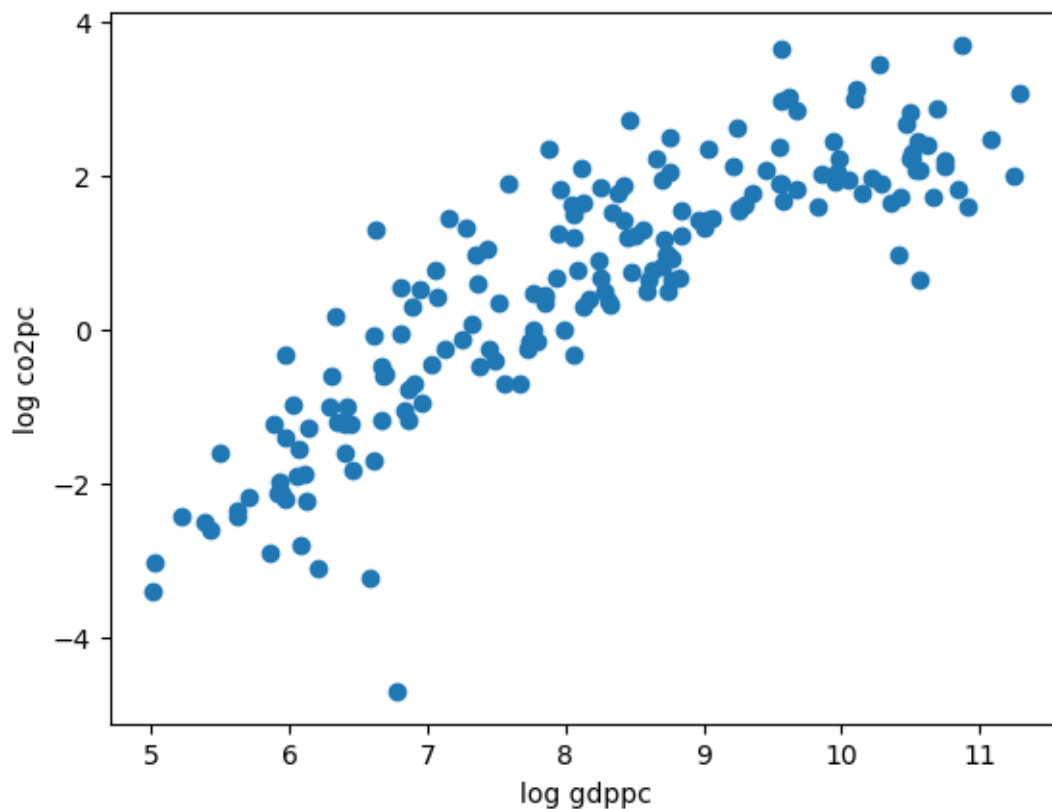


Question 4.f. Some countries with relatively high GDP might argue that their large population size – rather than carbon-intensive technology – drives high emissions. As a quick test on this claim, draw a scatterplot with the growth rates of per-capita GDP on the x-axis and the growth rates of per-capita emissions on the y-axis. You will need to generate these new variables/columns. To generate the plot you can mimic the code above but with your new variables in the right parts. Compare the resulting figure with the previous figure. Do you think the claim of the high-GDP countries is convincing? Explain.

This question is for your code, the next is for your explanation.

```
[25]: pollution['log_gdppc']=np.log(pollution['gdppc'])
pollution['log_co2pc']=np.log(pollution['co2pc'])

plt.scatter(pollution['log_gdppc'],pollution['log_co2pc'])
plt.xlabel("log gdppc")
plt.ylabel("log co2pc");
```



Question 4.g. Explain.

Yes, I do. It implies that as countries become wealthier on a per-person basis, their emissions per person rise as well.

1.5 Problem 5. Wages in Los Angeles

The dataset for this question contains information on several indicators for about 800 workers in the Los Angeles metropolitan area: hourly wages (**wage**), years of schooling (**education**), gender (**female**), years of work experience (**exp**), US citizen status (**citizen**), African-American status (**black**), and Hispanic origin (**hispanic**). The sample is taken from the 1990 Census of the Population.

```
[20]: la = pd.read_csv("la.csv")
      la.head()
      #800 workers
```

```
[20]:
```

	hispanic	citizen	black	exp	wage	female	education
0	1	1	0	14.0	5.288462	1	9
1	0	1	0	14.7	8.461538	1	13
2	0	1	0	14.7	10.416667	1	13

3	0	1	0	14.0	21.634615	1	14
4	1	0	0	12.0	3.365385	1	12

Question 5.a. Display summary statistics for this dataset that include at least the mean and interquartile range for each variable. You learned a command earlier in this assignment that does this.

```
[23]: summary_stats=la.describe()
      iqr=summary_stats.loc['75%']-summary_stats.loc['25%']
      summary_stats.loc['iqr']=iqr
      print(summary_stats)
```

	hispanic	citizen	black	exp	wage	female \
count	863.000000	863.000000	863.000000	863.000000	863.000000	863.000000
mean	0.367323	0.717265	0.088065	12.494044	13.413861	0.424102
std	0.482355	0.450590	0.283554	1.605983	14.659279	0.494492
min	0.000000	0.000000	0.000000	10.000000	1.250000	0.000000
25%	0.000000	0.000000	0.000000	11.000000	6.500000	0.000000
50%	0.000000	1.000000	0.000000	12.700000	10.576923	0.000000
75%	1.000000	1.000000	0.000000	14.000000	15.723952	1.000000
max	1.000000	1.000000	1.000000	15.000000	250.661540	1.000000
iqr	1.000000	1.000000	0.000000	3.000000	9.223952	1.000000

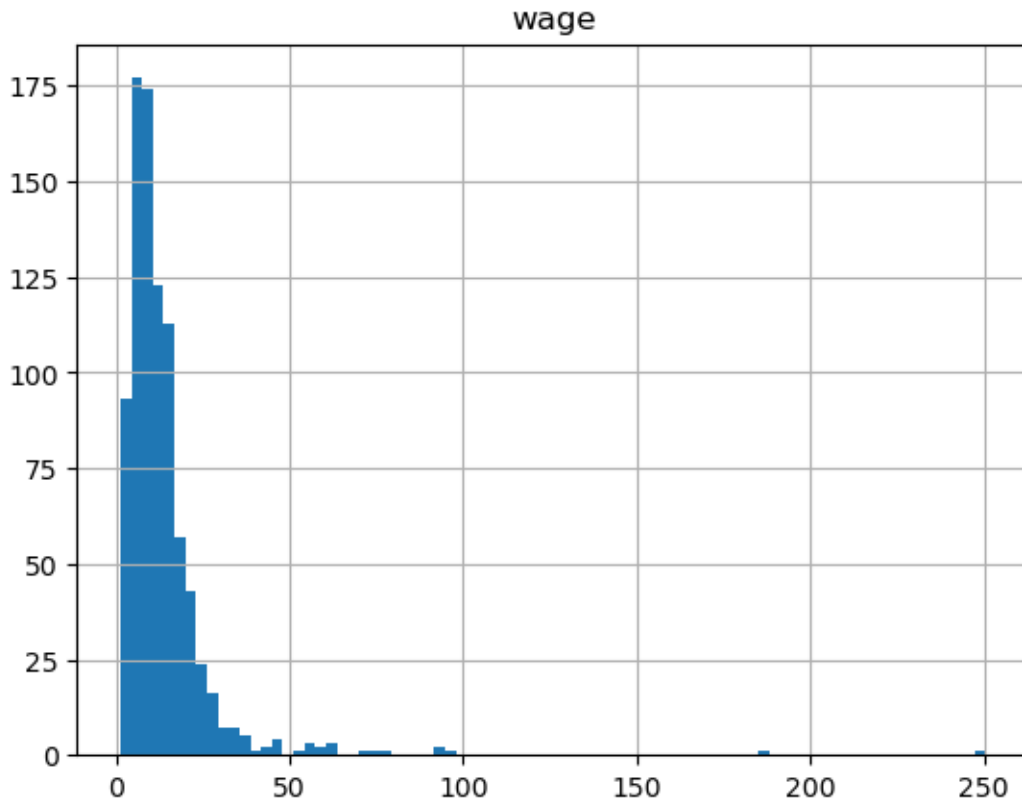
	education
count	863.000000
mean	12.864426
std	3.194241
min	0.000000
25%	12.000000
50%	13.000000
75%	16.000000
max	20.000000
iqr	4.000000

Question 5.b. Display a histogram *with 80 bins* of the wage variable. The histogram must have 80 bins. [This](#) is the documentation for the `.hist()` command for `pandas`, which may be helpful. The documentation for the `pandas` library is excellent and you are highly encouraged to reference it throughout the course.

Hint: If you wanted to generate a 40-bin histogram for the `exp` variable, the command would look like `la.hist(column='exp', bins=40)`.

```
[24]: la.hist(column='wage',bins=80)
```

```
[24]: array([[<Axes: title={'center': 'wage'}>]], dtype=object)
```



```
[27]: # This code cell is for question 5.c.
skewness=la['wage'].skew()
print("skewness of wage:", skewness)
```

skewness of wage: 8.186265111249103

Question 5.c. Is the distribution skewed in any way? You don't have to code for this part if you don't want to, a qualitative description is enough, but you can check quantitatively by doing `la.skew()`. If you do want to try this, use the code cell provided.

It is right skewed, actually, only a small number of employees can get high wages.

Question 5.d. Wages and earnings are often studied by first taking logarithms. Transform wage by taking its natural logarithm.

```
[29]: la['log_wage'] = np.log(la['wage'])

la.head()
```

```
[29]:   hispanic  citizen  black  exp    wage  female  education  log_wage
0         1         1     0  14.0   5.288462         1          9   1.665527
1         0         1     0  14.7   8.461538         1         13   2.135531
2         0         1     0  14.7  10.416667         1         13   2.343407
```

3	0	1	0	14.0	21.634615	1	14	3.074295
4	1	0	0	12.0	3.365385	1	12	1.213542

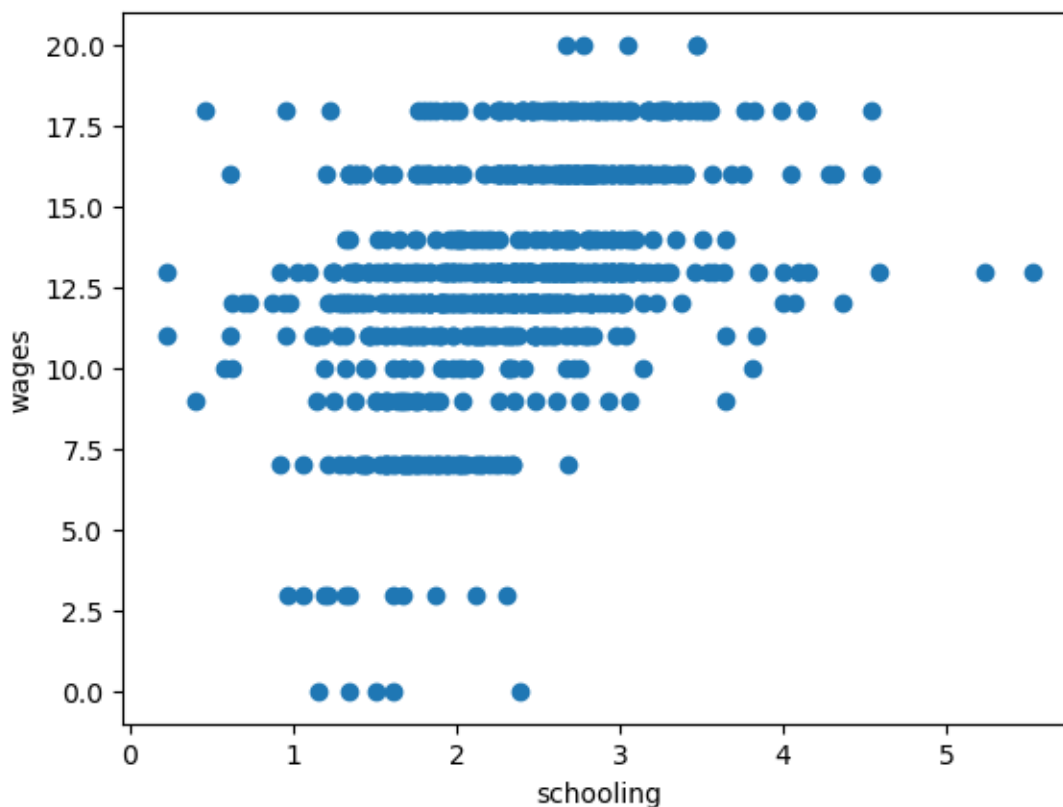
Question 5.e. Compute the frequencies of years of schooling (education). The `.value_counts()` method will be useful. [Here](#) is the documentation. For the `subset` parameter, you will want to set that equal to `'education'` (with the quote marks on either side; if you don't know why we need the quote marks, it might be worthwhile revisiting the Python assignment). You will also probably want to set the `sort` parameter equal to `False` because otherwise it will sort by count instead of education level, which might not be as helpful in this context. Set `normalize` to `True`. This will give proportions instead of counts. You don't have to do specify anything for the `ascending` parameter, it will default to `False` if you don't tell `pandas` what to do there.

```
[35]: edu_freq=la['education'].value_counts(normalize=True)
edu_freq
```

```
[35]: education
13    0.224797
12    0.201622
16    0.179606
18    0.086906
11    0.086906
7     0.068366
14    0.063731
9     0.033604
10    0.030127
3     0.012746
20    0.005794
0     0.005794
Name: proportion, dtype: float64
```

Question 5.f. Construct a scatterplot of log wages and years of schooling, with schooling on the x-axis and wages on the y-axis. Refer to how we used `plt.scatter()` earlier in the problem set if you don't recall how to make a scatterplot. Make sure to label your axes!

```
[40]: plt.scatter(la['log_wage'],la['education'])
plt.xlabel("schooling")
plt.ylabel("wages");
```



1.6 Problem 6. Sample mean and sample variance

Suppose we have an iid sample of size n of observations of the random variable Y , where $E[Y_i] = \mu$, $E[(Y_i - \mu)^2] = \sigma^2$, and $E[Y_i^4] < \infty$, for $i = 1, \dots, n$. Consider the quantities

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

$$S_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

In this exercise, we will show that the sample mean $\bar{Y}$ and the sample variance S_Y^2 are *unbiased* and *consistent* estimators of μ and σ^2 , respectively.

For this problem, you may either: * Write your math in the cells below using LaTeX typesetting (recommended). * Write your math on a tablet and include the exported pdf with the pdf submission of this assignment. * Write your math on paper and include a scan with the pdf submission of this assignment.

Question 6.a. Show that $\bar{Y}$ is an unbiased estimator of μ , i.e.,

$$E[\bar{Y}] = \mu$$

The sample mean is defined as $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$, and we need to prove that its expected value is equal to μ .

$$\begin{aligned} E[\bar{Y}] &= E \left[\frac{1}{n} \sum_{i=1}^n Y_i \right] \\ E[\bar{Y}] &= \frac{1}{n} E \left[\sum_{i=1}^n Y_i \right] \\ E[\bar{Y}] &= \frac{1}{n} \sum_{i=1}^n E[Y_i] \end{aligned}$$

It is an i.i.d. sample, and for each observation Y_i , its expected value is $E[Y_i] = \mu$. Put into the equation,

$$E[\bar{Y}] = \frac{1}{n} \sum_{i=1}^n \mu$$

The sum of n identical terms of μ is simply $n\mu$:

$$\begin{aligned} E[\bar{Y}] &= \frac{1}{n} (n\mu) \\ E[\bar{Y}] &= \mu \end{aligned}$$

So $\bar{Y}$ is an unbiased estimator of the population mean μ .

Question 6.b. Show that $\bar{Y}$ is a consistent estimator of μ , i.e.,

$$\text{plim } \bar{Y} = \mu$$

We need to justify it converges in probability to the true population parameter as the sample size n approaches infinity, $\text{plim}_{n \rightarrow \infty} \bar{Y} = \mu$.

Using the Weak Law of Large Numbers (WLLN), the observations $Y_1, Y_2, \dots, Y_n$ are i.i.d., with an expected value of $E[Y_i] = \mu$. According to WLLN, the sample mean $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$ will converge in probability to the expected value of each Y_i .

Therefore,

$$\text{plim}_{n \rightarrow \infty} \bar{Y} = E[Y_i]$$

Since $E[Y_i] = \mu$:

$$\text{plim}_{n \rightarrow \infty} \bar{Y} = \mu$$

Question 6.c. Let $Z_i = (Y_i - \bar{Y})$. Consider the quantity

$$\overline{Z^2} = \frac{1}{n} \sum_{i=1}^n Z_i^2$$

Show that

$$E[\overline{Z^2}] = \frac{n-1}{n} \sigma^2$$

Proof: $Z_i = (Y_i - \bar{Y})$, and $Z^2 = \frac{1}{n} \sum_{i=1}^n Z_i^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2$. This is the sample variance with a denominator of n .

$$(Y_i - \bar{Y}) = (Y_i - \mu) - (\bar{Y} - \mu)$$

Squaring:

$$(Y_i - \bar{Y})^2 = [(Y_i - \mu) - (\bar{Y} - \mu)]^2 = (Y_i - \mu)^2 - 2(Y_i - \mu)(\bar{Y} - \mu) + (\bar{Y} - \mu)^2$$

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (Y_i - \mu)^2 - 2(\bar{Y} - \mu) \sum_{i=1}^n (Y_i - \mu) + \sum_{i=1}^n (\bar{Y} - \mu)^2$$

The term $\sum_{i=1}^n (Y_i - \mu)$ can be rewritten:

$$\sum_{i=1}^n (Y_i - \mu) = \sum_{i=1}^n Y_i - n\mu = n\bar{Y} - n\mu = n(\bar{Y} - \mu)$$

Substitute this back into the sum:

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (Y_i - \mu)^2 - 2(\bar{Y} - \mu)[n(\bar{Y} - \mu)] + n(\bar{Y} - \mu)^2$$

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (Y_i - \mu)^2 - 2n(\bar{Y} - \mu)^2 + n(\bar{Y} - \mu)^2$$

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (Y_i - \mu)^2 - n(\bar{Y} - \mu)^2$$

And take the expectation of both sides:

$$E \left[\sum_{i=1}^n (Y_i - \bar{Y})^2 \right] = E \left[\sum_{i=1}^n (Y_i - \mu)^2 \right] - nE[(\bar{Y} - \mu)^2]$$

The term $E[(Y_i - \mu)^2]$ is, by definition, the variance of Y_i , which is σ^2 . The term $E[(\bar{Y} - \mu)^2]$ is the variance of the sample mean, which is $\text{Var}(\bar{Y}) = \frac{\sigma^2}{n}$.

$$E \left[\sum_{i=1}^n (Y_i - \bar{Y})^2 \right] = \sum_{i=1}^n E[(Y_i - \mu)^2] - n\text{Var}(\bar{Y})$$

$$E \left[\sum_{i=1}^n (Y_i - \bar{Y})^2 \right] = \sum_{i=1}^n \sigma^2 - n \left(\frac{\sigma^2}{n} \right) = n\sigma^2 - \sigma^2 = (n-1)\sigma^2$$

So, the expected value of Z^2 :

$$E[Z^2] = E \left[\frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2 \right] = \frac{1}{n} E \left[\sum_{i=1}^n (Y_i - \bar{Y})^2 \right] = \frac{1}{n} (n-1)\sigma^2$$

$$E[Z^2] = \frac{n-1}{n} \sigma^2$$

Question 6.d. From part (c), we see that $\overline{Z^2}$ is a *biased* estimator of σ . Show that $\overline{Z^2}$ is nevertheless a consistent estimator of σ^2 .

Proof: $E[Z^2] = \frac{n-1}{n}\sigma^2$. Since $E[Z^2] \neq \sigma^2$, Z^2 is a **biased estimator** of σ^2 . i.e., $\text{plim}_{n \rightarrow \infty} Z^2 = \sigma^2$. $Z^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2$ is a sample average. Use the WLLN, $E\left[\frac{1}{n} \sum_{i=1}^n (Y_i - \mu)^2\right] = \frac{1}{n} \sum_{i=1}^n E[(Y_i - \mu)^2] = \frac{1}{n} \sum_{i=1}^n \sigma^2 = \sigma^2$. And, $(Y_i - \mu)^2$ terms are i.i.d. with a mean of σ^2 , so we have:

$$\text{plim}_{n \rightarrow \infty} \left(\frac{1}{n} \sum_{i=1}^n (Y_i - \mu)^2 \right) = \sigma^2$$

Therefore,

$$\begin{aligned} Z^2 &= \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2 = \frac{1}{n} \sum_{i=1}^n [(Y_i - \mu) - (\bar{Y} - \mu)]^2 \\ Z^2 &= \frac{1}{n} \left[\sum_{i=1}^n (Y_i - \mu)^2 - n(\bar{Y} - \mu)^2 \right] = \frac{1}{n} \sum_{i=1}^n (Y_i - \mu)^2 - (\bar{Y} - \mu)^2 \\ \text{plim}_{n \rightarrow \infty} Z^2 &= \text{plim}_{n \rightarrow \infty} \left(\frac{1}{n} \sum_{i=1}^n (Y_i - \mu)^2 - (\bar{Y} - \mu)^2 \right) \end{aligned}$$

Using the property that the probability limit of a sum is the sum of the probability limits:

$$\text{plim}_{n \rightarrow \infty} Z^2 = \text{plim}_{n \rightarrow \infty} \left(\frac{1}{n} \sum_{i=1}^n (Y_i - \mu)^2 \right) - \text{plim}_{n \rightarrow \infty} (\bar{Y} - \mu)^2$$

As we know from part 6.b, $\text{plim}_{n \rightarrow \infty} \bar{Y} = \mu$. And put it into the equation:

$$\text{plim}_{n \rightarrow \infty} (\bar{Y} - \mu)^2 = (\text{plim}_{n \rightarrow \infty} \bar{Y} - \mu)^2 = (\mu - \mu)^2 = 0$$

Therefore:

$$\text{plim}_{n \rightarrow \infty} Z^2 = \sigma^2 - 0 = \sigma^2$$

Question 6.e. Finally, show that S_Y^2 is an unbiased and consistent estimator of σ^2 .

Proof: The sample variance S_Y^2 is defined as $S_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$. From part 6.c, $E[\sum_{i=1}^n (Y_i - \bar{Y})^2] = (n-1)\sigma^2$. So,

$$E[S_Y^2] = E\left[\frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2\right] = \frac{1}{n-1} E\left[\sum_{i=1}^n (Y_i - \bar{Y})^2\right] = \frac{1}{n-1} (n-1)\sigma^2 = \sigma^2$$

Since $E[S_Y^2] = \sigma^2$, the sample variance S_Y^2 is an unbiased estimator of σ^2 .

As for consistency, we must show that $\text{plim}_{n \rightarrow \infty} S_Y^2 = \sigma^2$.

As for transformation, $S_Y^2 = \frac{n}{n-1} Z^2$. From part 6.d, $\text{plim}_{n \rightarrow \infty} Z^2 = \sigma^2$, $\text{plim}_{n \rightarrow \infty} \frac{n}{n-1} = 1$, we have:

$$\text{plim}_{n \rightarrow \infty} S_Y^2 = \text{plim}_{n \rightarrow \infty} \left(\frac{n}{n-1} Z^2 \right) = \left(\text{plim}_{n \rightarrow \infty} \frac{n}{n-1} \right) \cdot (\text{plim}_{n \rightarrow \infty} Z^2) = (1)(\sigma^2) = \sigma^2$$

$\text{plim}_{n \rightarrow \infty} S_Y^2 = \sigma^2$, so the sample variance S_Y^2 is a consistent estimator of σ^2 .

1.7 Submission

Make sure you have run all cells in your notebook in order before exporting. **Please save before exporting.**